Aref Afzali

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Summary — Software and ML Engineer (M.Sc. CS) who enjoys building real products end-to-end, from data and models to reliable backend services. I've worked on LLM-based systems, search and recommendation workflows, and production deployments, and I like turning messy problems into simple experiences for users.

Experience

***Co-Founder & ML Engineer* | Qliqless.ai | Montreal, QC, Canada** **Apr 2025 – Oct 2025**

- Built and shipped an AI conversational agent MVP that connects natural-language requests to real business actions via tool endpoints.
- Designed a **multi-tenant** architecture using **MCP** to provision **per-business tool endpoints** and customer-specific agent configurations.
- Containerized and deployed services with **Docker/Kubernetes** on **AWS (EC2)** behind an **NGINX reverse proxy**, enabling reproducible environments and scalable rollout.
- Delivered multiple end-to-end demos to pilot users and iterated quickly based on feedback, translating product requirements into weekly engineering deliverables.

***R&D Engineer* | XRaise.ai | San Francisco, CA, USA (Remote)** **Dec 2024 – Feb 2025**

- Built a **RAG-based recommendation system** using **LlamaIndex** + **Qdrant** to match startups with similar YC/TechStars companies; integrated **Exa Search API** for real-time company data retrieval.
- Implemented a **hybrid retrieval router** combining **vector similarity** (Qdrant) with **keyword-based retrieval** (LlamaIndex) to improve coverage across sparse and exact-match queries.
- Evaluated retrieval quality using **LlamaIndex** RAG evaluation (**DeepEval**, **RAGAS**) to compare retrieval strategies and tune top-k.
- Architected an LLM-assisted pipeline to extract structured fields from **VC and grant application forms** by combining HTML analysis with **multimodal agents** (Browser-Use), and **LLM-powered processing** (HyperBrowser).

***Software Team Lead* | RomaParvaz Travel Agency | Tehran, Iran** **Mar 2022 – Mar 2023**

- Led end-to-end development of an airline ticketing platform from design to production using Scrum; coordinated delivery across backend, frontend, and integrations.
- Architected a **microservices** and **event-driven** backend (NestJS) with React frontend and PostgreSQL/MongoDB/Redis to support scalable booking workflows.
- Integrated multiple GDS providers (**Amadeus, TravelPort, Gabriel**) and built resilient handling for third-party latency/failures (timeouts, retries, fallbacks).
- Owned production delivery: containerized services with **Docker** and **Docker Compose**; implemented **GitLab CI** pipelines (and Jenkins for legacy jobs) for automated builds and releases.
- Configured **NGINX** as reverse proxy with **TLS**, **rate limiting**, and **caching**; used **Cloudflare** for DNS/CDN; implemented centralized logging with the **ELK stack** (Elasticsearch + Logstash + Kibana) for monitoring and debugging.

***Data Specialist* | Carriot Company | Tehran, Iran** **Jul 2020 – Apr 2021**

- Built a **Vehicle Routing Problem (VRP) optimization API** in **Flask** using **Google OR-Tools** and custom **heuristic/metaheuristic** methods to support PDPTW, time windows, open depots, and multi-depot routing.
- Designed routing objectives and feasibility constraints; tuned solver/heuristic settings to balance distance, timing, and operational requirements.
- Developed a **Persian address geocoding service** using **Sent2Vec** embeddings and clustering to improve address resolution for logistics workflows.
- Implemented **vehicle behavior analytics** (stop-type detection, acceleration-axis calibration) using speed/acceleration + geospatial signals with data preprocessing and feature engineering.
- Built **geospatial heatmap visualizations** to communicate demand patterns and fleet movement insights to stakeholders.

Education

Concordia University **Sep 2024 - Aug 2026**
Master of Science in Computer Science (CGPA: 3.9/4)
Supervisor: Prof. Hovhannes Harutyunyan
M.Sc. Thesis on Broadcasting Algorithms on K-Path Graphs

University of Tehran **Sep 2017 - Jul 2022**
Bachelor of Science in Engineering Science (Software Engineering Branch) (CGPA: 16.20/20)
B.Sc. Thesis on an Application of Basket Analysis Using Data Mining Approaches
Minors: Computer Science *Sep 2019 - Jul 2022*

Publications

- Afzali A., Bashizade M., Akbarein H. (2023). **The Application of Spiking Neural Network in Schizophrenia.** *1st International Congress of Artificial Intelligence in Medical Sciences (AIMS 2023).* (A Poster Presentation)
- Shayegh B., Afzali A., MohammadHashemi S., MohammadTaheri K., Mohammadi S. (2021). **Discrete Mathematics: An Introduction with an Academic Approach.** *Open-source book manuscript* (link available upon request)

Projects

Complex Networks

- Multimodal Graph-Based Recommendation System
  - Implemented MGCLTransformer, integrating **Graph Transformers** with **Multimodal Graph Contrastive Learning (MGCL)** to process user-item interactions based on item images and titles. Achieved 21.4% improvement in Hit Ratio and 44.1% in NDCG over baseline models. Used PyTorch, PyTorch Geometric, SentenceTransformer, and ResNet-50 for implementation.
- Graph Analysis & Community Detection
  - Analyzed **centrality measures** (Eigenvector Centrality, Betweenness Centrality, etc.) across multiple datasets.
  - Applied **community detection algorithms** (Spectral Clustering, K-Means) and **dimensionality reduction** (PCA, Laplacian Eigenmaps).
  - Developed **fuzzy community detection models** using C-means and **Non-negative Matrix Factorization** (NMF).
  - Utilized **Graph Convolutional Networks** (GCNs) for image classification, demonstrating the effectiveness of graph-based deep learning.

Business Intelligence

- Basket Recommender Application (B.Sc. Thesis)
  - Developed a recommendation system using **Association Rules algorithms** to suggest products and target customers based on purchase history. Implemented the model with Flask-RESTful (backend), VueJS (frontend), and PostgreSQL (database), and further integrated its outputs into **Power BI** for visualization and performance analytics. Achieved **2nd place** among B.Sc. theses selected by the faculty.

Machine Learning & Artificial Intelligence

- Music Genre Classification
  - Worked on the Neuromatch Academy (2022) project for classifying music genres using machine learning on spectral and temporal audio features.
- Optimized CNN for Efficient Computation
  - Designed a CNN model with CUDA, leveraging float16 precision for improved efficiency on the MNIST dataset.
- Text Classification & Decoding Optimization
  - Built a news classification system using Bayesian Networks for probabilistic inference.
  - Developed a genetic algorithm-based replacement decoding method to optimize text reconstruction.

Skills

ML/LLM PyTorch, LlamaIndex, scikit-learn, RAG, embeddings, Claude/ChatGPT tool calling
Data Python (NumPy, Pandas), NetworkX, Matplotlib
Back-end Django, Flask, REST APIs, NestJS, Java, C++
Infra Docker, Docker Compose, Kubernetes, NGINX, GitLab CI, Linux/Shell
Cloud AWS (EC2), Cloudflare (DNS, CDN, TLS)
Databases PostgreSQL, MySQL, MongoDB, Redis, Qdrant, Elasticsearch
Tools Git, DVC
Familiar Spark, Hadoop, Neo4j, Cassandra, Scala, Go, Jenkins

Honors & Achievements

- Ranked in **top 1%** of the National University Entrance Examination, earning **a full tuition waiver** for B.Sc. studies; achieved **2nd rank** for undergraduate thesis and 5th rank overall in the program.
- Successfully participated in prestigious AI research programs, including **Neuromatch Academy Deep Learning** and **ANITI's Reinforcement Learning Virtual School**.
- Qualified for the first stage of both National Mathematics and Computer Olympiads; completed competitive phases of multiple Cognitive Neuroscience Competitions.

Languages

- English (Advanced)
- Persian (Native)
- French (Elementary)
- German (Elementary)